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Laporan Akhir

Praktikum Jaringan Komputer

VLAN, Trunk, OSPF MultiVendor, dan Implementasi Jaringan Enterprise

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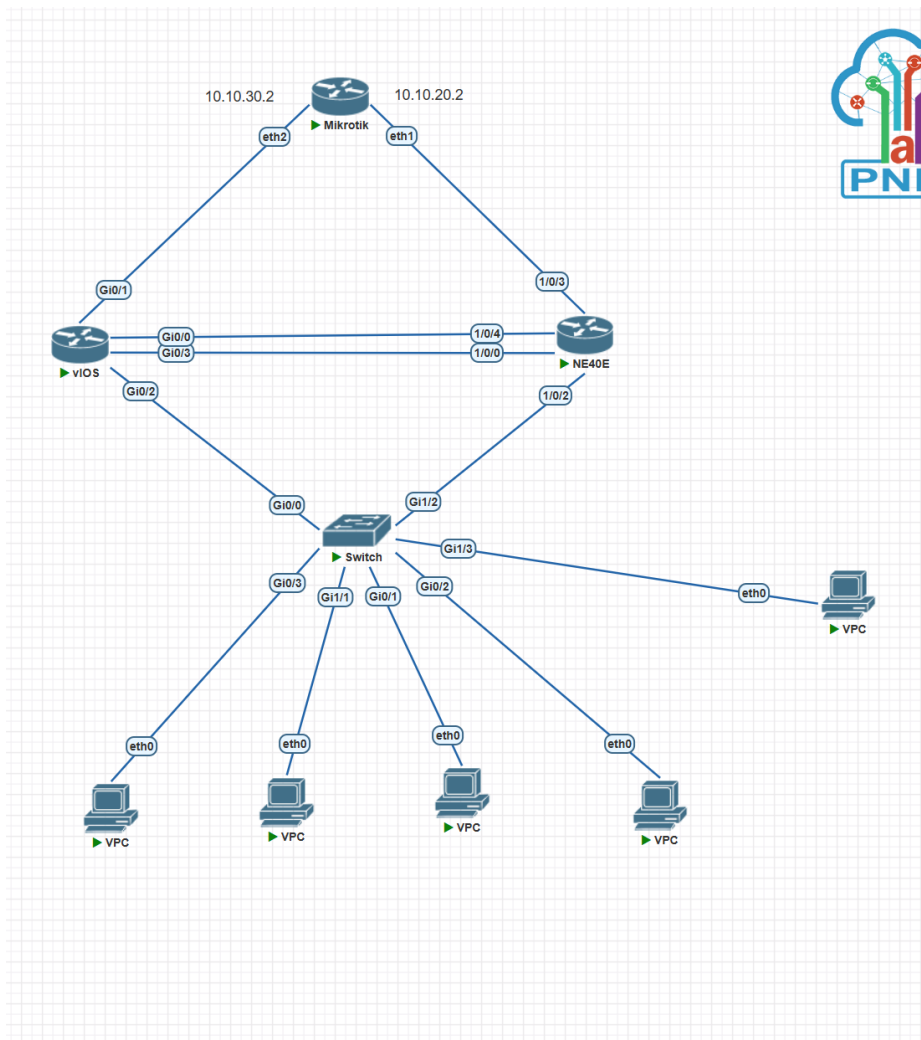
2026

1 Langkah-Langkah Percobaan

Praktikum Modul 5 membahas konfigurasi jaringan berbasis VLAN, trunk, routing antarperangkat, OSPF multivendor, serta pengujian failover. Topologi utama menggunakan Cisco Switch, Cisco Router, Huawei Router, MikroTik Router, dan beberapa VPCS sebagai end-device. Selain praktikum utama, tugas modul mengembangkan topologi menjadi jaringan enterprise HQ–Branch yang melibatkan VRRP, DHCP relay, FortiGate, GRE tunnel, dan OSPF.

1.1 Persiapan Topologi dan Addressing

Langkah awal adalah menyusun topologi sesuai referensi Modul 5. Pada praktikum utama, VLAN yang digunakan adalah VLAN 10, VLAN 20, VLAN 30, dan VLAN 40 dengan gateway masing-masing pada jaringan 192.168.10.0/24, 192.168.20.0/24, 192.168.30.0/24, dan 192.168.40.0/24. Link antar-router menggunakan network /30 agar tiap hubungan point-to-point memiliki alamat yang efisien.



Gambar 1: Topologi praktikum Modul 5 VLAN, trunk, dan OSPF multivendor.

1.2 Konfigurasi Cisco Switch

Pada Cisco Switch dibuat VLAN 10, 20, 30, dan 40. Port yang menuju VPCS dikonfigurasi sebagai access port sesuai VLAN masing-masing, sedangkan port yang menuju Cisco Router dikonfigurasi

sebagai trunk port dengan VLAN allowed 10, 20, 30, dan 40. Konfigurasi ini diperlukan agar beberapa VLAN dapat melewati satu link fisik menuju router.

```
Switch#show vlan
VLAN Name                Status Ports
-----
1    default              active Gi0/0, Gi1/0, Gi1/2, Gi1/3
10   Finance              active Gi0/2
20   HR                  active Gi0/1
30   IT                  active Gi1/1
40   Marketing            active Gi0/3
1002 fddi-default          act/unsup
1003 token-ring-default  act/unsup
1004 fddinet-default     act/unsup
1005 trnet-default       act/unsup
Switch#
```

Gambar 2: Verifikasi VLAN pada Cisco Switch.

```
Switch#show interfaces trunk
Port      Mode          Encapsulation  Status        Native vlan
Gi0/0     on            802.1q         trunking      1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     none
Switch#
```

Gambar 3: Verifikasi trunk pada Cisco Switch.

1.3 Konfigurasi Cisco Router

Cisco Router dikonfigurasi dengan konsep router-on-a-stick menggunakan subinterface pada interface trunk ke switch. Setiap subinterface diberi encapsulation dot1Q sesuai VLAN dan alamat gateway. VLAN 10 dan VLAN 20 menggunakan DHCP pool dari Cisco Router, sedangkan VLAN 30 dan VLAN 40 menggunakan IP static pada VPCS. Router juga dikonfigurasi untuk link point-to-point menuju Huawei dan MikroTik.

```

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    10.20.20.2      YES NVRAM  up          up
GigabitEthernet0/1    192.168.10.1    YES NVRAM  up          up
GigabitEthernet0/2    unassigned      YES NVRAM  administratively down down
GigabitEthernet0/3    unassigned      YES NVRAM  administratively down down
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.20.20.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.20.20.1
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      10.20.20.0/30 is directly connected, GigabitEthernet0/0
L      10.20.20.2/32 is directly connected, GigabitEthernet0/0
      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.10.0/24 is directly connected, GigabitEthernet0/1
L      192.168.10.1/32 is directly connected, GigabitEthernet0/1
Router#show running-config
Building configuration...

Current configuration : 2931 bytes
!
version 15.9
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname Router
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
Router#

```

Gambar 4: Konfigurasi interface dan subinterface pada Cisco Router.

```

Router#show ip dhcp pool
Pool VLAN10-FINANCE :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)          : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 0
  Pending event                     : none
  1 subnet is currently in the pool :
    Current index      IP address range      Leased addresses
    192.168.10.1       192.168.10.1 - 192.168.10.254  0
Router#
Pool VLAN20-HR :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)          : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 0
  Pending event                     : none
  1 subnet is currently in the pool :
    Current index      IP address range      Leased addresses
    192.168.20.1       192.168.20.1 - 192.168.20.254  0
Router#

```

Gambar 5: Konfigurasi DHCP pool pada Cisco Router.

1.4 Konfigurasi Huawei Router dan MikroTik Router

Huawei Router dikonfigurasi sebagai salah satu router pada jaringan multivendor. Interface Huawei diberi alamat IP untuk link ke Cisco, MikroTik, dan LAN tambahan. MikroTik Router juga diberi IP

pada interface ke Cisco dan Huawei sehingga seluruh router membentuk jalur redundan. Setelah IP antarrouter selesai, konektivitas dasar diuji menggunakan ping sebelum masuk ke konfigurasi OSPF.

Interface	IP Address/Mask	Physical	Protocol	VPN
Ethernet1/0/0	10.10.10.2/30	up	up	--
Ethernet1/0/0.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/1	unassigned	up	down	--
Ethernet1/0/1.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/2	192.168.50.1/24	up	up	--
Ethernet1/0/2.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/3	10.10.20.1/30	up	up	--
Ethernet1/0/3.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/4	10.10.11.2/30	up	up	--
Ethernet1/0/4.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/5	unassigned	up	down	--
Ethernet1/0/5.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/6	unassigned	up	down	--
Ethernet1/0/6.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/7	unassigned	up	down	--
Ethernet1/0/7.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/8	unassigned	up	down	--
Ethernet1/0/8.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/9	unassigned	up	down	--
Ethernet1/0/9.4094	128.1.138.137/16	up	up	l3vpn
GigabitEthernet0/0/0	unassigned	up	down	--
LoopBack2147483647	128.1.138.137/16	up	up(s)	l3vpn
VLL0	unassigned	up	up(s)	--

Gambar 6: Konfigurasi interface pada Huawei Router.

#	ADDRESS	NETWORK	INTERFACE
0	;; TO-HUAWEI 10.10.20.2/30	10.10.20.0	ether1
1	;; TO-CISCO 10.10.30.2/30	10.10.30.0	ether2

Gambar 7: Konfigurasi alamat IP pada MikroTik Router.

1.5 Konfigurasi OSPF Multivendor

OSPF dikonfigurasi pada Cisco, Huawei, dan MikroTik untuk mengiklankan network VLAN, LAN, dan link antarrouter. Area yang digunakan adalah area backbone agar seluruh router dapat saling bertukar informasi link-state. Setelah OSPF aktif, dilakukan verifikasi neighbor dan routing table pada setiap perangkat untuk memastikan adjacency terbentuk.

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	1	FULL/BDR	00:00:33	10.10.30.2	GigabitEthernet0/1
2.2.2.2	1	FULL/BDR	00:00:31	10.10.11.2	GigabitEthernet0/0
2.2.2.2	1	FULL/BDR	00:00:32	10.10.10.2	GigabitEthernet0/3

Gambar 8: OSPF neighbor pada Cisco Router.

```
[~HUAWEI]display ospf peer
(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
  State: Full           Mode:Nbr is Slave   Priority: 1
  DR: 10.10.10.1       BDR: 10.10.10.2     MTU: 1500
  Dead timer due in 37 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m09s
  Neighbor Up Time : 2026-06-06 04:07:43
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
  State: Full           Mode:Nbr is Slave   Priority: 1
  DR: 10.10.11.1       BDR: 10.10.11.2     MTU: 1500
  Dead timer due in 37 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m09s
  Neighbor Up Time : 2026-06-06 04:07:43
  Authentication Sequence: [ 0 ]
```

Gambar 9: OSPF peer pada Huawei Router.

```
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=10.10.20.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
  adjacency=3m5s
1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
  adjacency=3m12s
[admin@MikroTik] > |
```

Gambar 10: OSPF neighbor pada MikroTik Router.

1.6 Pengujian Failover OSPF

Pengujian failover dilakukan dengan mematikan salah satu atau dua link pada jalur Cisco–Huawei. Tujuannya adalah melihat apakah routing table berubah dan apakah paket tetap dapat mencapai tujuan melalui jalur alternatif. Pengujian dilakukan menggunakan ping dan traceroute dari VPCS ke network tujuan.

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.950 ms 5.009 ms 3.982 ms
 2 10.10.11.2 7.390 ms 4.217 ms 5.330 ms
 3 *192.168.50.10 11.236 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS> |
```

Gambar 11: Traceroute menuju network 192.168.50.0/24 pada kondisi normal.

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 4.724 ms 4.092 ms 3.875 ms
 2 10.10.11.2 6.939 ms 5.072 ms 10.426 ms
 3 *192.168.50.10 6.885 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 12: Traceroute setelah salah satu link mengalami perubahan jalur.

2 Analisis Hasil Percobaan

Hasil konfigurasi menunjukkan bahwa VLAN pada switch berhasil dibuat dan port trunk berhasil membawa VLAN 10, 20, 30, dan 40. Hal ini dibuktikan melalui output `show vlan brief` dan `show`

interfaces trunk. Access port membuat host hanya berada pada satu VLAN tertentu, sedangkan trunk memungkinkan frame dari beberapa VLAN melewati satu koneksi menuju router.

Pada Cisco Router, subinterface dot1Q berhasil menjadi gateway antar-VLAN. DHCP pool juga berjalan karena client pada VLAN DHCP memperoleh alamat IP sesuai jaringan masing-masing. Untuk VLAN static, VPCS dapat berkomunikasi ketika IP address dan gateway dikonfigurasi sesuai tabel addressing. Dengan demikian, konsep router-on-a-stick terbukti dapat melakukan inter-VLAN routing melalui satu interface fisik.

Konfigurasi OSPF pada Cisco, Huawei, dan MikroTik berhasil membentuk neighbor antarrouter. Routing table pada perangkat menunjukkan rute OSPF menuju network yang tidak terhubung langsung. Hal ini menunjukkan bahwa perbedaan vendor tidak menghalangi pertukaran rute selama parameter OSPF seperti network, area, dan interface telah sesuai.

Pengujian failover menunjukkan bahwa OSPF mampu memilih jalur alternatif ketika salah satu link mengalami gangguan. Perubahan jalur dapat diamati dari hasil traceroute dan tabel routing. Pada kondisi tertentu, delay sementara dapat terjadi karena router membutuhkan waktu untuk mendeteksi perubahan neighbor, memperbarui LSDB, lalu menghitung ulang rute terbaik menggunakan algoritma SPF.

```
Router#show ip ospf neighbor
Neighbor ID    Pri   State           Dead Time   Address        Interface
3.3.3.3        1     FULL/BDR        00:00:34    10.10.30.2     GigabitEthernet0/1
2.2.2.2        1     FULL/BDR        00:00:33    10.10.11.2     GigabitEthernet0/0
2.2.2.2        1     FULL/BDR        00:00:29    10.10.10.2     GigabitEthernet0/3
Router#show ip ospf rib

      OSPF Router with ID (1.1.1.1) (Process ID 1)

      Base Topology (MTID 0)

OSPF local RIB
Codes: * - Best, > - Installed in global RIB

* 10.10.10.0/30, Intra, cost 10, area 0, Connected
  via 10.10.10.1, GigabitEthernet0/3
* 10.10.11.0/30, Intra, cost 10, area 0, Connected
  via 10.10.11.1, GigabitEthernet0/0
*> 10.10.20.0/30, Intra, cost 110, area 0
  via 10.10.10.2, GigabitEthernet0/3
  via 10.10.11.2, GigabitEthernet0/0
* 10.10.30.0/30, Intra, cost 100, area 0, Connected
  via 10.10.30.1, GigabitEthernet0/1
* 192.168.10.0/24, Intra, cost 1, area 0, Connected
  via 192.168.10.1, GigabitEthernet0/2.10
* 192.168.20.0/24, Intra, cost 1, area 0, Connected
  via 192.168.20.1, GigabitEthernet0/2.20
* 192.168.30.0/24, Intra, cost 1, area 0, Connected
```

Gambar 13: Routing table OSPF pada Cisco Router.

```

[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
Public_ Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask  Proto  Pre  Cost    Flags NextHop         Interface
-----
10.10.30.0/30    OSPF   10   110     D   10.10.11.1   Ethernet1/0/4
10.10.30.0/30    OSPF   10   110     D   10.10.10.1   Ethernet1/0/0
192.168.10.0/24  OSPF   10   11      D   10.10.11.1   Ethernet1/0/4
192.168.10.0/24  OSPF   10   11      D   10.10.10.1   Ethernet1/0/0
192.168.20.0/24  OSPF   10   11      D   10.10.11.1   Ethernet1/0/4
192.168.20.0/24  OSPF   10   11      D   10.10.10.1   Ethernet1/0/0
192.168.30.0/24  OSPF   10   11      D   10.10.11.1   Ethernet1/0/4
192.168.30.0/24  OSPF   10   11      D   10.10.10.1   Ethernet1/0/0
192.168.40.0/24  OSPF   10   11      D   10.10.11.1   Ethernet1/0/4
192.168.40.0/24  OSPF   10   11      D   10.10.10.1   Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

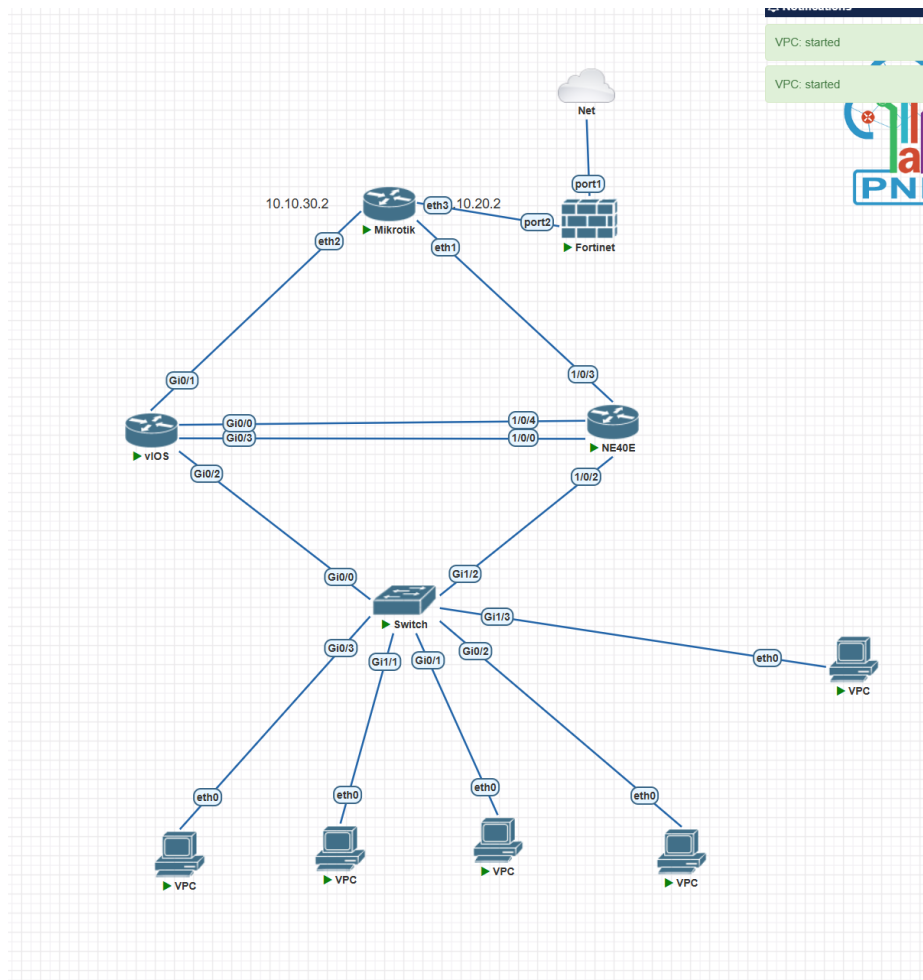
Destination/Mask  Proto  Pre  Cost    Flags NextHop         Interface
-----
10.10.10.0/30    OSPF   10   10      10.10.10.2   Ethernet1/0/0
10.10.11.0/30    OSPF   10   10      10.10.11.2   Ethernet1/0/4
10.10.20.0/30    OSPF   10  100     10.10.20.1   Ethernet1/0/3
192.168.50.0/24  OSPF   10   1       192.168.50.1 Ethernet1/0/2
[~HUAWEI]

```

Gambar 14: Routing table OSPF pada Huawei Router.

3 Hasil Tugas Modul

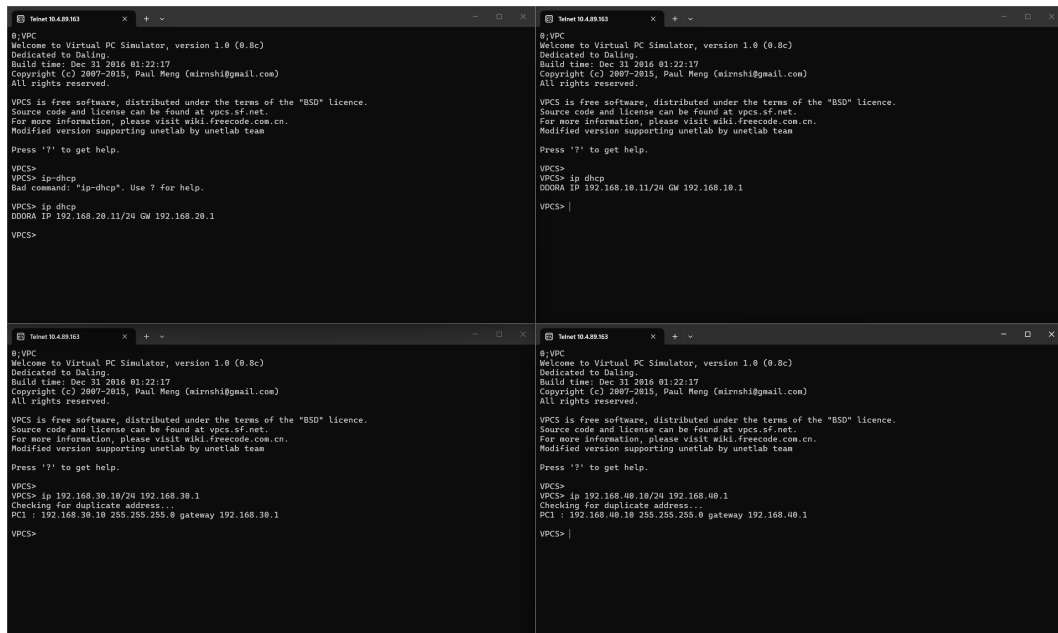
Tugas modul mengimplementasikan jaringan enterprise yang menghubungkan HQ Jakarta dan Branch Surabaya. Topologi menggunakan VLAN pada sisi Jakarta, VRRP sebagai gateway redundan, Ubuntu Server sebagai DHCP dan web server, FortiGate sebagai firewall dan endpoint GRE tunnel, MikroTik sebagai router ISP, serta OSPF untuk pertukaran rute antar-site.



Gambar 15: Topologi tugas modul enterprise HQ–Branch.

3.1 VLAN, Trunk, dan Router Jakarta

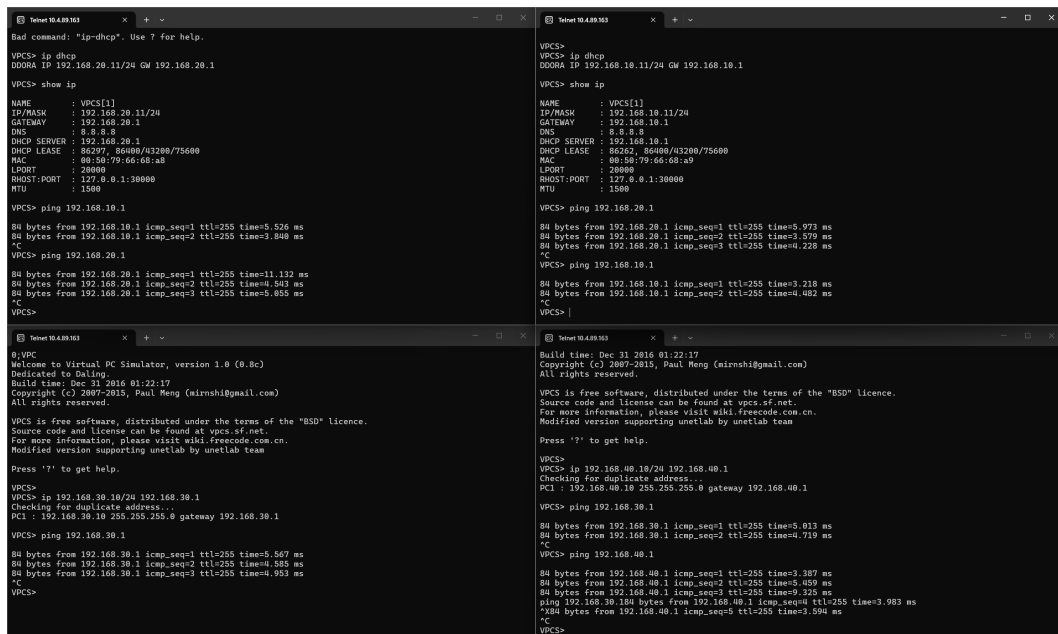
Pada sisi Jakarta, Cisco Switch dikonfigurasi untuk membawa VLAN 10, VLAN 20, dan VLAN 60. VLAN 10 digunakan untuk Finance, VLAN 20 untuk IT, dan VLAN 60 untuk server HQ. Cisco Router Jakarta dan MikroTik Jakarta menjalankan VRRP sehingga gateway virtual tetap tersedia apabila salah satu router mengalami gangguan.



Gambar 16: Konfigurasi dan perolehan IP pada client Jakarta.

3.2 DHCP Server dan Web Server Jakarta

Ubuntu Server Jakarta berada pada VLAN 60 dengan alamat 192.168.60.10/24. Server ini berperan sebagai ISC-DHCP Server untuk VLAN 10 dan VLAN 20 serta sebagai web server menggunakan Nginx. DHCP relay pada router meneruskan request client menuju Ubuntu Server sehingga client tetap dapat memperoleh IP meskipun server berada pada VLAN berbeda.



Gambar 17: Pengujian koneksi client terhadap gateway dan jaringan lain.

3.3 FortiGate, GRE Tunnel, dan OSPF

FortiGate Jakarta dan FortiGate Surabaya dikonfigurasi sebagai firewall sekaligus endpoint GRE tunnel. GRE tunnel menghubungkan dua site melalui jaringan ISP sehingga traffic antar-kantor dapat

lewat melalui interface tunnel. OSPF kemudian digunakan untuk mengiklankan network Jakarta dan Surabaya melalui jalur tunnel tersebut.

```
FortiGate-VM64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4
       * - candidate default

Routing table for VRF=0
S*   0.0.0.0/0 [10/0] via 10.10.10.1, port1, [1/0]
C    10.10.10.0/30 is directly connected, port1
C    10.20.20.0/30 is directly connected, port2
S    192.168.10.0/24 [10/0] via 10.20.20.2, port2, [1/0]
C    192.168.20.0/24 is directly connected, port3

FortiGate-VM64-KVM # get system interface physical
== [onboard]
    ==[port1]
        mode: static
        ip: 10.10.10.2 255.255.255.252
        ipv6: ::/0
        status: up
        speed: 10000Mbps (Duplex: full)
        FEC: none
        FEC_cap: none
    ==[port2]
        mode: static
        ip: 10.20.20.1 255.255.255.252
        ipv6: ::/0
        status: up
        speed: 10000Mbps (Duplex: full)
        FEC: none
        FEC_cap: none
    ==[port3]
        mode: static
        ip: 192.168.20.1 255.255.255.0
        ipv6: ::/0
        status: up
        speed: 10000Mbps (Duplex: full)
        FEC: none
        FEC_cap: none
    ==[port4]
        mode: static
        ip: 0.0.0.0 0.0.0.0
        ipv6: ::/0
        status: down
        speed: n/a
        FEC: none
        FEC_cap: none
```

Gambar 18: Konfigurasi FortiGate pada tugas modul.

```
FortiGate-VM64-KVM #
FortiGate-VM64-KVM # get router info routing-table all

command parse error before 'ingo'
Command fail. Return code -61

FortiGate-VM64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4
       * - candidate default

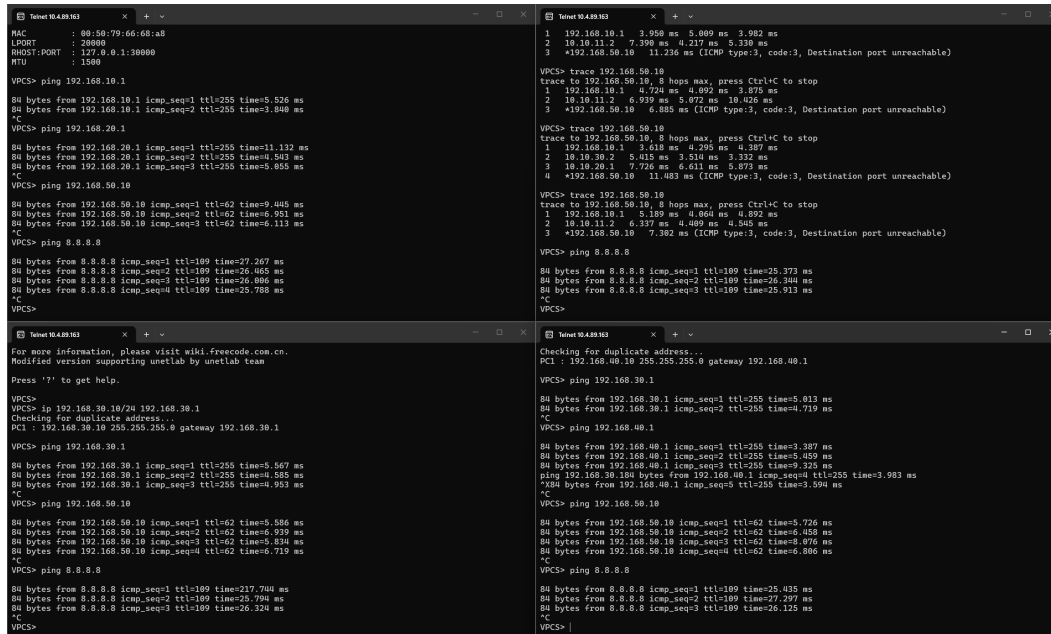
Routing table for VRF=0
S*   0.0.0.0/0 [5/0] via 10.0.137.1, port1, [1/0]
C    10.0.137.0/24 is directly connected, port1
C    10.10.40.0/30 is directly connected, port2
S    192.168.0.0/16 [10/0] via 10.10.40.1, port2, [1/0]

FortiGate-VM64-KVM # |
```

Gambar 19: Routing table pada FortiGate.

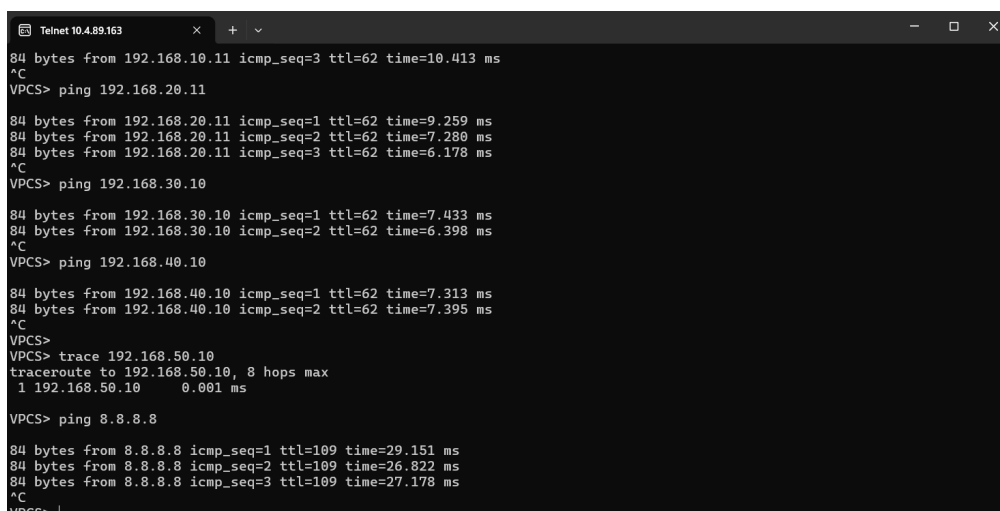
3.4 Pengujian Konektivitas End-to-End

Pengujian dilakukan dari client Jakarta, client Surabaya, dan perangkat WAN. Hasil ping dan trace-route menunjukkan bahwa client dapat mencapai gateway, jaringan remote, dan alamat pengujian internet. Dengan adanya OSPF, rute antar-site dapat dipelajari secara dinamis, sedangkan GRE tunnel menyediakan jalur logis antarkantor.



The image shows four terminal windows from a Telnet session on a Cisco router (Telnet 10.4.89.163). The first window shows the configuration of interfaces 10/24, 10/20, and 10/10, and the execution of ping commands to 192.168.10.1, 192.168.20.1, and 192.168.50.10. The second window shows the execution of ping commands to 8.8.8.8 and 192.168.50.10. The third window shows the execution of ping commands to 192.168.30.1, 192.168.40.1, and 192.168.50.10. The fourth window shows the execution of ping commands to 8.8.8.8 and 192.168.50.10, and the execution of the trace command to 192.168.50.10.

Gambar 20: Pengujian ping dari VPCS ke alamat tujuan.



The image shows a terminal window from a Telnet session on a Cisco router (Telnet 10.4.89.163). The terminal displays the results of ping commands to 192.168.20.11, 192.168.30.10, 192.168.40.10, and 8.8.8.8. It also shows the execution of the trace command to 192.168.50.10, which displays the path taken by the traffic.

Gambar 21: Pengujian koneksi WAN pada tugas modul.

4 Kesimpulan

Berdasarkan praktikum Modul 5, VLAN dapat digunakan untuk memisahkan jaringan secara logis pada satu switch fisik, sedangkan trunk memungkinkan beberapa VLAN dikirimkan melalui satu link. Router-on-a-stick pada Cisco Router berhasil menyediakan inter-VLAN routing dan DHCP untuk jaringan client. OSPF multivendor berhasil membentuk adjacency antara Cisco, Huawei, dan MikroTik

serta mendistribusikan rute secara dinamis. Pada tugas modul, konsep tersebut diperluas menjadi jaringan enterprise HQ–Branch dengan VRRP, DHCP relay, FortiGate, GRE tunnel, dan OSPF sehingga jaringan memiliki gateway redundan, layanan DHCP terpusat, serta konektivitas antarkantor.

5 Lampiran

5.1 Dokumentasi saat praktikum

Dokumentasi berikut menunjukkan bukti tambahan konfigurasi dan pengujian selama praktikum serta pengerjaan tugas modul.

```
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.a9    Jun 07 2026 03:34 AM    Automatic
192.168.20.11    0100.5079.6668.a8    Jun 07 2026 03:34 AM    Automatic
Router#
Router#
```

Gambar 22: DHCP binding pada Cisco Router.

```
[admin@MikroTik] > /routing ospf neighbor print
0 instance-default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=0.0.0.0 state="Full" state-changes=5 ls-retransmits=1 ls-requests=0 db-summaries=0 adjacency=4s
1 instance-default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
  adjacency=11s
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 ADo 10.10.10.0/30      10.10.30.1    110
  10.10.20.1
1 ADo 10.10.11.0/30       10.10.30.1    110
  10.10.20.1
2 ADo 192.168.10.0/24       10.10.30.1    110
3 ADo 192.168.20.0/24       10.10.30.1    110
4 ADo 192.168.30.0/24       10.10.30.1    110
5 ADo 192.168.40.0/24       10.10.30.1    110
6 ADo 192.168.50.0/24       10.10.20.1    110
[admin@MikroTik] >
```

Gambar 23: Konfigurasi OSPF pada MikroTik.

5.2 Hasil Challenge Modul

Challenge modul dikerjakan dengan melakukan pengujian jalur dan konektivitas tambahan pada topologi enterprise. Hasil traceroute menunjukkan jalur yang dilewati paket saat menuju network tujuan, sedangkan konfigurasi manual route dan firewall digunakan sebagai bagian dari penyesuaian agar traffic dapat melewati perangkat yang sesuai.

```
Telnet 10.4.89.163
* Technical Advisory Center. Any use or disclosure, in whole or in part, *
* of the IOSv Software or Documentation to any third party for any *
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****
Router>
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#
Router(config)#ip access-list standard PBR-VLAN30-40
Router(config-std-nacl)#
Router(config-std-nacl)# permit 192.168.30.0 0.0.0.255
Router(config-std-nacl)#
Router(config-std-nacl)# permit 192.168.40.0 0.0.0.255
Router(config-std-nacl)#
Router(config-std-nacl)#exit
Router(config)#route-map VIA-HUAWEI permit 10
Router(config-route-map)#
Router(config-route-map)# match ip address PBR-VLAN30-40
Router(config-route-map)#
Router(config-route-map)# set ip next-hop 10.10.10.2
Router(config-route-map)#
Router(config-route-map)#exit
Router(config)#interface gig0/2.30
Router(config-subif)#
Router(config-subif)# ip policy route-map VIA-HUAWEI
Router(config-subif)#
Router(config-subif)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#interface gig0/2.40
Router(config-subif)#
Router(config-subif)# ip policy route-map VIA-HUAWEI
Router(config-subif)#
Router(config-subif)#exit
Router(config)#
```

Gambar 24: Konfigurasi manual route pada challenge modul.

```

FortiGate-VM64-KVM # show firewall policy
config firewall policy
  edit 1
    set name "LAN_to_WAN"
    set uuid 262db69c-5fa2-51f1-f787-56aa5a879390
    set srcintf "port2"
    set dstintf "port1"
    set action accept
    set srcaddr "LAN_Network"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
    set nat enable
  next
  edit 2
    set name "LAN_to_DMZ"
    set uuid 41805846-5fa2-51f1-dd27-b463cdb83971
    set srcintf "port2"
    set dstintf "port3"
    set action accept
    set srcaddr "LAN_Network"
    set dstaddr "Server_DMZ"
    set schedule "always"
    set service "ALL"
  next
  edit 3
    set name "WAN_to_DMZ_HTTP"
    set uuid 7958000c-5fa2-51f1-4123-e1f5b6bca40a
    set srcintf "port1"
    set dstintf "port3"
    set action accept
    set srcaddr "Client_WAN"
    set dstaddr "VIP_DMZ_Web"
    set schedule "always"
    set service "HTTP"
  next
end

FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM # show firewall vip
config firewall vip
  edit "VIP_DMZ_Web"
    set uuid ffd8e96c-5fa1-51f1-4714-9b23cd423dff
    set extip 10.10.10.2
    set mappedip "192.168.20.10"
    set extintf "port1"
    set portforward enable
    set extport 80
    set mappedport 80
  next
end

```

Gambar 25: Konfigurasi firewall dan VIP pada FortiGate untuk challenge.